Part 1: Multiple Choice

1. Why does transforming ATP to ADP release stored energy for cellular work?
   A) By absorbing a photon of light in the chloroplast
   B) By breaking an unstable molecule into more stable products
   C) By combining with glucose to form carbon dioxide
   D) By combining individual molecules to a more stable final product

2. What are the primary reactants required for photosynthesis to take place in a plant?
   A) Oxygen, glucose, and water
   B) Carbon dioxide, ATP, and chlorophyll
   C) Glucose, sunlight, and water
   D) Carbon dioxide, water, and sunlight

3. Which cellular molecule and structure gives plant leaves their green color?
   A) Chlorophyll inside mitochondria
   B) ATP synthase inside mitochondria
   C) Chlorophyl inside chloroplasts
   D) ATP synthase inside chloroplasts

4. What is the primary purpose of the Calvin cycle in photosynthesis?
   A) To split water molecules and release oxygen gas
   B) To absorb sunlight using chlorophyll
   C) To use energy carriers (ATP and NADPH) and carbon dioxide to build glucose
   D) To break down glucose into ATP for immediate use

5. Which of the following equations accurately summarizes aerobic cellular respiration?
   A) Carbon dioxide + Water + Sunlight → Glucose + Oxygen
   B) Glucose + Oxygen → Carbon dioxide + Water + ATP
   C) Glucose + Carbon dioxide → Lactic acid + ATP
   D) Water + Oxygen → Glucose + ATP

6. In which cellular organelle does most of aerobic cellular respiration occur?
   A) Mitochondria
   B) Chloroplast
   C) Ribosome
   D) Golgi apparatus

7. Which description is accurate for the direction electrons are carried in cellular respiration and photosynthesis
   A) In respiration and photosynthesis electrons: glucose → ETC
   B) In respiration: glucose → ETC, and in photosynthesis: ETC → glucose
   C) In respiration and photosynthesis: ETC → glucose
   D) In respiration: ETC → glucose, and in photosynthesis: glucose → ETC

8. Which initial stage of cellular respiration happens in the cytoplasm and is anaerobic?
   A) The Calvin cycle
   B) The electron transport chain
   C) The Krebs cycle
   D) Glycolysis

9. What type of anaerobic process occurs in human muscle cells during extreme, fast-paced exercise when oxygen levels become low?
   A) Alcoholic fermentation
   B) The Krebs cycle
   C) Lactic acid fermentation
   D) Photosynthesis

10. How are photosynthesis and cellular respiration interdependent in an ecosystem?
   A) They both consume glucose and release sunlight
   B) The products of one process serve as the reactants for the other process
   C) They occur within the exact same organelle in plant cells
   D) They both require external supplies of ATP to begin

11. How does the total ATP yield from aerobic respiration compare to that of anaerobic fermentation for a single glucose molecule?

A) Aerobic respiration produces a much larger amount of ATP than fermentation

B) Fermentation produces far more ATP because it bypasses the mitochondria

C) Both processes yield the exact same amount of ATP per glucose molecule

D) Fermentation produces ATP, whereas aerobic respiration consumes ATP without producing any net gain

12. What molecules are the electron *acceptors* in cellular respiration vs. photosynthesis

A) Oxygen is the electron respiration, and water in photosynthesis

B) NAD^+ is the acceptor in respiration and $NADP^+$ in photosynthesis

C) NAD^+ is the acceptor in respiration, and oxygen in photosynthesis

D) Oxygen is the acceptor in respiration, and $NADP^+$ in photosynthesis

13. Where does the Krebs cycle occur?

A) Chloroplast stroma

B) Mitochondrial matrix

C) Inner mitochondrial membrane

D) Thylakoid membrane

14. Why do photosynthetic organisms use the energy from sunlight to make large stores of ATP, rather than immediately consuming the ATP by making glucose?

A) Turning the glucose back into CO_2 gets the cells a higher yield ATP than directly turning the sunlight into ATP only

B) Glucose made in the Calvin Cycle is not used for energy, but to build the cell. Directly made ATP from the ETC is used for energy in other cellular processes.

C) ATP is too unstable a molecule to be used for long term storage, and cells need an energy source even when sunlight is unavailable

D) Cells cannot make ATP directly from sunlight – the energy goes directly into building glucose, and only from there to ATP

Part 2: Free Response

15. A student places a healthy green plant inside a sealed glass container with plenty of water and rich soil. The container is left under a bright lamp.
    a. Describe what happens to the levels of carbon dioxide and oxygen inside the container during this time.

    b. After 1 week, the student replaces the plant with several pill bugs (along with an adequate food source) Describe what will happen to the carbon dioxide and oxygen levels following this change.

16. During a 100-meter dash, an athlete's muscles quickly run out of oxygen, yet they are able to finish the race.
    a. Identify the anaerobic pathway the muscle cells switch to when oxygen is scarce.

    b. Identify the substance that builds up in the muscle cells, causing the "burn" feeling.

    c. Which substance would be produced if a yeast cell engaged this pathway?

    d. Explain why cells engage this pathway, if it does not directly produce additional energy?

17. Imagine you seal a snail and a water plant together inside a jar filled with water and place it near a sunny window. Both organisms manage to survive for months without any outside air or food.
    a. Explain how the chemical outputs of the snail support the survival of the plant.

    b. Explain how the chemical outputs of the plant support the survival of the snail.

    c. Write out or describe the balanced relationship between their primary energy processes.

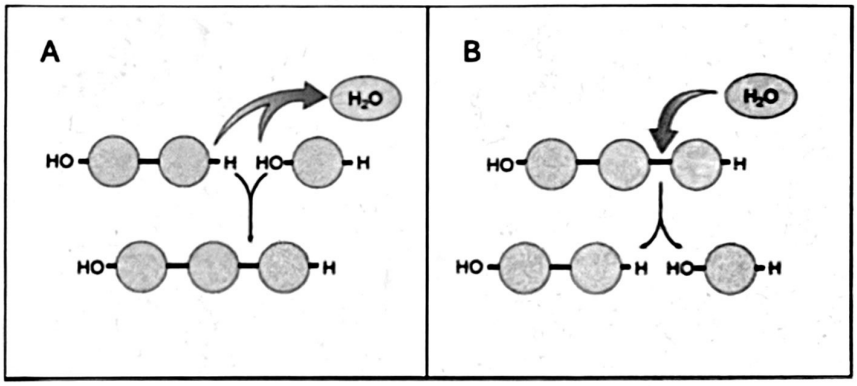
18. Explain the difference between autotrophy and heterotrophy.

    a. Do autotrophs, heterotrophs, or both, utilize photosynthesis?

    b. Do autotrophs, heterotrophs, or both, utilize cellular respiration?

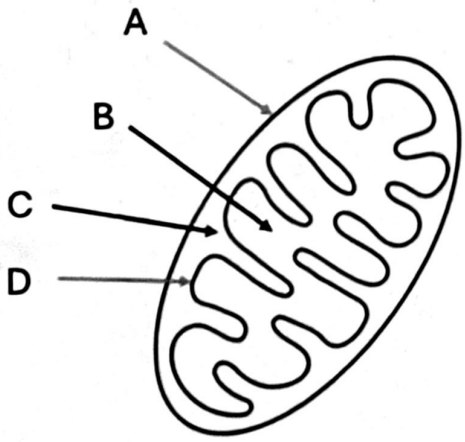
    c. Describe two different ways heterotrophs can absorb nutrients

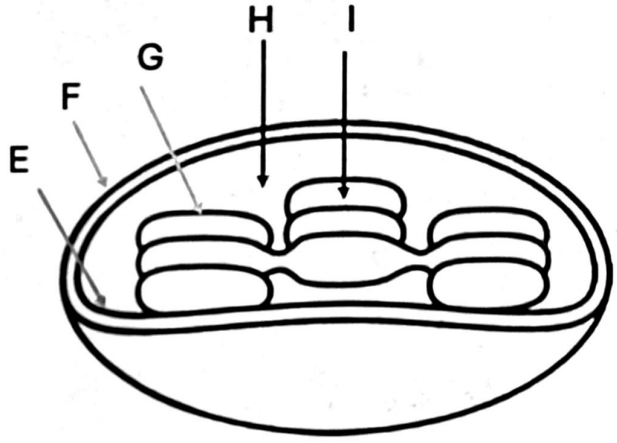
19. Which of the following reactions likely releases energy, and which likely requires an energy input? Explain why



20. Identify the components labelled in the following diagrams

    Note – orange arrows are pointing to membranes





21. Identify where the following processes create a high concentration of protons

    a) ETC in aerobic respiration of animal cells
    b) ETC in photosynthesis
    c) ETC of oxidative phosphorylation in bacteria

22. Identify the molecule through which protons flow back down their concentration gradient, producing ATP

Answer Key

Multiple Choice

1. B	2. C	3. C	4. B	5. B	6. A	7. B
8. D	9. C	10. B	11. A	12. D	13. B	14. C

Free Response

15. CO_2 levels will decrease, O_2 levels will increase.

After exchanging the plant for pill bugs, CO_2 levels will increase and O_2 will decrease

16. a. fermentation

b. lactic acid

c. alcohol

d. it recycles NADH back to NAD^+, allowing glycolysis to continue producing ATP

17. The plant produces oxygen, which is necessary for the snail to engage in aerobic respiration. The snail produces CO_2 gas, which is necessary for the plant to engage in photosynthesis.
The plant may also produce glucose for the snail, when leaves or other bits of plant matter fall off and get consumed.

18. Autotrophs use energy to convert small molecules into big molecules, building their own food, which then serves as a stable energy source as needed.
Heterotrophs obtain energy from breaking down large molecules they encounter or absorb.

a. Autotrophs utilize photosynthesis

b. Both types of organisms utilize cellular respiration

c. Heterotrophs can absorb large complex nutrient sources whole (eating or engulfing them), and then break them down internally OR, they can excrete enzymes to break the nutrients down, and absorb the smaller resulting molecules.

19. Process A likely requires an energy input. It is combining smaller molecules into a big one; This generally means the products are less energetically stable, and always means they are less entropically favorable (it increases order in the universe).

Process B likely produces energy; it is breaking a larger molecule into smaller ones. This generally means the products are more energetically stable, and always means they are entropically favorable (it increases disorder in the universe)

20. A. Outer mitochondrial membrane
B. Mitochondrial membrane
C. Inter-membrane space (mitochondria)
D. Inner mitochondrial membrane
E. Inner chloroplast membrane
F. Outer chloroplast membrane
G. Thylakoid membrane
H. Chloroplast stroma
I. Thylakoid space

21. a) in the intermembrane space of mitochondria
b) in the thylakoid space, inside chloroplasts
c) in between their cell membrane and their cell wall

22. ATP Synthase